Alexander H. Zhong

617-755-9998 | alexhyzhong@gmail.com | linkedin.com/in/alexander-zhong-331b79173 | alexanderzhong.com

Experience

Google Mountain View, CA

Software Engineer, Google Payments / Google Wallet *June 2022 – Present*

GPay Brazil / Pix *Mar. 2024 – Present*

- Designed and implemented features in the Java backend Orchestration layer for Pix in Google Wallet, coordinating business logic and data flows across client, backend, and partner systems supporting 31 banks, 1.5M monthly transacting users, and 6M users with a linked form of payment
- Led Orchestration design and implementation for a Pix bank-balance pilot, building balance APIs and data loaders, defining protobuf/RPC contracts, and designing consent and cross-service flows for sensitive financial data
- Reduced authentication-flow drop-off by 2–3 percentage points from a 12–13% baseline across approximately 500K users through error handling and fallback improvements
- Designed and implemented end-to-end telemetry and automated alerting for FIDO authentication and Play Integrity in Pix payment flows operating at a 96% success rate; built A/B dashboards that validated an NFC checkout optimization, reducing latency by 2–3 seconds and increasing conversion by 3.4%

GPay US Features / GPay North America *June 2022 – Mar. 2024*

- Owned end-to-end design and implementation of recurring transaction management using server-driven UI across Flutter client, backend aggregation, and persisted user state
- Rolled the feature out to 330K monthly active users, reaching 23% unique-user click-through; 14-day app retention reached 56%, 8 percentage points above baseline
- Implemented server-authoritative age controls for approximately 1M users, translating legal requirements into eligibility and DOB remediation flows; 33% regained access within two weeks

Google Mountain View, CA

Software Engineering Intern, Google Payments Autofill *June 2021 – Aug. 2021*

- Developed a C++ feature-provider module in a distributed data aggregation framework to serve real-time account and security risk signals for Virtual Card Number enrollment and retrieval
- Aggregated account and security-history signals for issuer risk assessment, enabling more than 65% of low-risk users to bypass additional CVC or OTP verification; launched with experiment-based rollout and unit testing

Projects

Wanderform.com | *React, Node.js, Express, Firebase, Firestore, Anthropic API* June 2025 – Present

- Designed and launched a full-stack AI travel planner with a React client, Express backend, Firebase Authentication, and Firestore persistence
- Built a multi-step Claude-powered pipeline that transforms free-form preferences into structured itineraries using structured outputs, validation, retries, and prompt-driven editing
- Built an AI edit-preview workflow that generates revised itineraries from user prompts and lets users compare changes before applying them

Video Game Popularity Prediction | *Python, PyTorch, NumPy, pandas, scikit-learn* Fall 2025

- Co-developed a 513-feature dataset of approx. 27K Steam games from game metadata and engagement signals
- Implemented and analyzed k-means and Gaussian Mixture Model clustering to identify interpretable game segments and evaluate clustering quality and stability

Education

Carnegie Mellon University Pittsburgh, PA

B.S. in Computer Science; Additional Major in Mathematical Sciences; GPA: 3.89/4.0; High Honors *May 2022*

Stanford University Palo Alto, CA

Graduate Certificate in Artificial Intelligence *Sep. 2025 – Present*

- Completed Coursework: Machine Learning (CS229), Natural Language Processing with Deep Learning (CS224N)

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, C++, Go, SQL
Frameworks and Tools: React, Node.js, Express, Firebase, Firestore, Flutter, gRPC, Protobuf, Git, REST
AI/ML: PyTorch, NumPy, pandas, scikit-learn, Anthropic API